

Junyi Li

Tel: 917-914-1134 | Email: jl14864@nyu.edu

EDUCATION

New York University, New York, USA
Ph.D. in Urban System

- GPA: 3.87/4.00
- Advisor: Prof. Eleonora Maria Tronci
- Relevant Course: Monitoring City, Urban Infrastructure Management

New York University, New York, USA
M.S. in Applied Urban Science and Informatics

May 2025

- GPA: 3.95/4.00
- Capstone Research Project: Data-driven modeling of depression in urban areas
- Relevant Course: Complex Urban System, Bayesian Machine Learning

Tongji University, Shanghai, CHN
BArch in Architecture (2024 QS #13)

July 2023

- GPA: 88.33/100
- Dissertation: Autonomous Optimization Design Method of Near-Zero Energy Buildings
- Relevant Course: Building Mechanics, Disaster Prevention for Building, Computer Graphics, Simulation Technology of Digital Landscape

RESEARCH INTERESTS

- Urban Infrastructure Resilience
- Reinforcement Learning
- Decision making

PUBLICATIONS

- [1] Li, J., Gan, T., Lyu, S., Wu, H., Macrì, S., & Porfiri, M. (2026). *The mechanisms beneath the urban buffer against depression*. *Cities*, 176, 107215.
- [2] Li, J., & Tronci, E. M. (2026). NOI-POMDP: A Noise-Explicit POMDP Framework for Structural Health Monitoring. In *Proceedings of the 12th European Workshop on Structural Health Monitoring (EWSHM 2026)*, Toulouse, France.
- [3] Li, J., Qiu, M., Yan, C., & Yuan, F. (2022). *Research on discrete construction design for cluster elastic deformation*. In *Proceedings of the 2022 National Academic Conference on Architectural Digital Technology* (pp. 281-286).

[4] Li, J., Zhang, Z., Mendel, T., & Yabe, T. *Evaluating Impact of NYC Sidewalk Sheds on Local Business Through AI-Chatbot Survey*. (Under Review)

HONORS | AWARDS | FELLOWSHIP

CUSP Experiential Scholars Scholarship (\$7,000)	2024
Tongji Foundation International Student Alumni Award (\$500)	2024
Outstanding graduates of Shanghai (Top 5% of graduates in Shanghai)	2023
Tongji University Academic Scholarship (\$1,000)	2022
Excellent Prize, Competition for the Design of Rural Primary School	2021
Grand Prize, Shanghai Social Practice Project Competition for College Students	2019

CONFERENCES

Presenter, Engineering Mechanics Institute Conference 2026 June 2026
Boulder, Colorado, USA

- Finalist of probabilistic-methods-committee-student-paper-competition
- Introduced a noise-aware POMDP framework for condition-based maintenance under time-varying sensor uncertainty.
- Improved long-term inspection and maintenance policies through adaptive observation modeling and online sensor reliability learning.

Presenter, 2025 American Association of Geographer Annual Meeting March 2024
Detroit, Mich, USA

- Developing an AI-Chatbot web page with large language models using Javascript and Python to collect public participation data.
- Conducting image-based analysis using CNN to identify features from user drawings; utilizing NLP-based methods to calibrate concept emergence from qualitative data.

PROFESSIONAL EXPERIENCE

Research Assistant Sept 2023 – Present
The Furman Center for Real Estate and Urban Policy, New York, USA

- **Data Processing:** Spearheaded the implementation of an ETL pipeline using AWS CLI, SQL, and R, improving data processing speed by 25% through streamlined data transformations. Designed Quality Control metrics and automated monthly HTML reports, saving 10 hours of manual work per month. Linked 85 MB eviction records with housing subsidized databases through a geocoding process, reducing data mismatch rates by 80%.
- **A/B Testing:** Developed control and test groups for a 2-year A/B test using the Kolmogorov-Smirnov test, ensuring accurate, reliable results.
- **Data Visualization:** Created data visualizations using Python, R, and Tableau, improving public understanding and supporting data requests.

- **Project management:** Managed data coding documentation, meeting notes, and task tracking for three projects; developed a documentation template adopted company-wide.

Computational Design Research Intern

ECADI Intelligent Design Research Center, Shanghai, CHN

Mar 2023 – June 2023

- **Software Development:** Developed a Python-based data platform to extract and exchange data between AEC applications, transforming 3D data into JSON, and enhancing interoperability. Spearheaded the development of a feature for Rhino and Autodesk Revit to automate MEP plan generation from residential building floor plans.
- **Optimized Codebase:** Modularized and optimized the 35 MB Code File to 20 MB, enhancing functionality and reducing dirty reads by 16.7%.

Research Intern

Guiyang Big Data Industry Co., Ltd., Guizhou, CHN

Dec. 2022 – Feb 2023

- **Literature Review:** Formulated a policy memo on building Green Data Centers and upgrading the nationwide power grid to support energy transition.
- **Webinar Organization:** Supported webinars on spatial-temporal big data integration across city departments, and negotiated partnerships with Guizhou University and Beijing Jiaotong University.

SKILLS

Programming Languages: Python, MySQL, R, C#, HTML, CSS, JavaScript

Software: MS Office, Adobe Creative Suite, AutoCAD, Revit, Rhino, Unity, ArcGIS, ESRI

Technical Skills: Machine Learning, Statistical Modeling, Causal Inference, Git workflow

Languages: English, Chinese

SERVICE

Student Organizations:

- **Director of Development and Partnership**, Applied Urban Science Association, New York University (2023.10 - 2025.5)
- **Leader of Planning and Operating**, Alumni Liaison Association, Tongji University (2018.9 - 2023.7)
- **Director of Student Development**, College of Architecture and Urban Planning (CAUP) Student Union, Tongji University (2019.9 - 2021.9)

Volunteer:

- **Communication Ambassador**, Guangzhou “Blue Envelope” Care Center for Left-behind Children (2020.5 - 2023.7)
- **Art Teacher**, Shanghai Fu Mei Social Welfare Development Center (2019.9 - 2021.12)